

Behrooz Farkiani

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Fields of Interest

- Networked Systems
- Applied Machine Learning and Optimization
- Holistic System Design and Implementation

Education

2022–2027 **Ph.D., Computer Science,**
Washington University in St. Louis, USA
2012–2014 **M.Sc., Information Technology Engineering – Computer Networks,**
Amirkabir University of Technology, Tehran, Iran
2007–2012 **B.Sc., Information Technology Engineering,**
Amirkabir University of Technology, Tehran, Iran

Notable Work Experience

2025–2026 Applied Scientist Internship (L5), Amazon, San Diego, USA
Improving business decision making by utilizing optimization and machine learning approaches.

2022–2027 Graduate Research Assistant, Washington University in St. Louis, USA
Lead architect and developer of the Hermes overlay, designed to improve service delivery over the internet.

2015–2020 Graduate Research Assistant, Amirkabir University, Tehran, Iran
Designed and prototyped an integrated SDN-NFV architecture, focusing on management components and creating algorithms to efficiently solve NP-hard problems.

2018–2019 Internship on Software Defined Networking, Ghent, Belgium
Designed and implemented a flow bandwidth allocation and routing algorithm to ensure QoS for high-priority flows in an SDN environment.

2017–2018 IT Consultant, Technical and Vocational University, Tehran, Iran
Consulted to streamline student-facing portals for a university with over 150,000 students, addressing critical issues caused by poorly designed Oracle databases.

Recent Publications (2025+)

- Behrooz Farkiani, Fan Liu, Patrick Crowley. *Enabling Network Policy Enforcement in Service Meshes*. 2026 IEEE/IFIP Network Operations and Management Symposium (NOMS), 2026.
- Fan Liu, Behrooz Farkiani, Patrick Crowley. *Time-Series Foundation Models for ISP Traffic Forecasting*. 2026 IEEE/IFIP Network Operations and Management Symposium (NOMS), 2026.
- Behrooz Farkiani, Fan Liu, Patrick Crowley. *Rethinking HTTP API Rate Limiting: A Client-Side Approach*. 2026 IEEE Consumer Communications and Networking Conference (CCNC), 2026.
- Fan Liu, Behrooz Farkiani, Patrick Crowley. *Large Language Models for Computer Networking Operations and Management: A Survey on Applications, Key Techniques, and Opportunities*. Computer Networks, vol. 271, p. 111614, Oct. 2025.
- Fan Liu, Behrooz Farkiani, John Dehart, Jyoti Parwatikar, Patrick Crowley. *Performance Comparison of HTTP/3 and HTTP/2 with Proxy Integration*. In Proceedings of the 34th International Conference on Computer Communications and Networks (ICCCN 2025), pages 1–6, 2025.

Notable Projects (Main Contributor)

- Toward Automated Network Policy Configuration (2025–)
 - Designing an intent-driven configuration framework for service meshes that uses SMT-based models to capture and formally verify network policies. Developer intent is expressed in natural language, translated by a large language model (LLM) into mesh-specific configuration, and iteratively refined with an SMT-based verifier to ensure correctness and alignment between intent and implementation. (Ongoing)
- Toward Formally Verifying Congestion Control Algorithms (2025)
 - Investigated formal verification of AIMD and BBR congestion control algorithms using SMT solvers to test performance and safety properties beyond simulations and small-scale emulations. Extended an existing model with refined implementation details and used the Z3 SMT solver to verify whether user-specified conditions can occur and under which parameters. Demonstrated that steady-state bounds reported in the original model do not hold under the refined model. [Project Page]
- Enabling Network Policy Enforcement in Service Meshes (2025)
 - Proposed a lightweight network overlay that integrates with service meshes (Istio, Linkerd, Consul) to enforce Layer 3 network policies independently of underlying infrastructure. The design decouples policy enforcement from cluster networking, addressing the finding that 91% of Kubernetes clusters do not effectively enforce network policies. The overlay enables service meshes to provide end-to-end Layer 3-7 policy enforcement in a portable, infrastructure-agnostic manner across heterogeneous environments. [Paper (IEEE/IFIP NOMS 2026)]
- Rethinking API Rate Limiting: A Client-Side Approach (2025)
 - Developed client-side rate-limiting mechanisms that reshape HTTP retry behavior without requiring service-provider changes. The approach addresses exponential retry

growth when multiple independent clients share an endpoint and use naive exponential backoff. The mechanisms infer congestion from implicit feedback (e.g., HTTP 429 responses), reduce 429 errors by 97%, and can be implemented in web browsers using service workers. [Project Page] [IEEE CCNC 2026]

- Hermes Overlay (2022–2025)
 - Designed Hermes, an end-to-end general-purpose overlay architecture to improve service delivery on the Internet. Hermes addresses four long-standing architectural limitations: lack of true end-to-end traffic management, poor backward compatibility for new service interfaces, fragmented data-plane security and privacy, and the absence of an adaptable communication layer that can be programmed according to service needs and network changes. Implemented core components including the overlay controller, infrastructure controller, observability tools, and an Android client using Go, Python, Lua, Kotlin, Bash, Ansible, Docker, Open Policy Agent, and Envoy Proxy. [Project Page] [Paper (under review)]
- Improving Theoretical Aspects of Service Function Chain Embedding
 - Studied theoretical aspects of service function orchestration in NFV environments, focusing on NP-hard resource allocation problems. Designed exact algorithms that reach optimal solutions up to 8 faster than IBM CPLEX. Developed heuristics that produce near-optimal solutions for very large instances in under 10 seconds. Built a discrete-event network simulator in Java (50k+ lines of code) to evaluate these algorithms. [IEEE/ACM TNET 2021] [IEEE TNSM 2019] [Elsevier FGCS 2019]
- Flow Prioritization and Bandwidth Allocation in SDN Environments
 - Completed a research internship at Ghent University (Belgium), where I designed an OpenFlow-based SDN mechanism to prioritize incoming flows and guarantee dedicated bandwidth for high-priority traffic. Modeled the system in IBM CPLEX to derive optimal policies, then implemented them on SDN switches using OpenFlow v1.3 to efficiently apply the theoretical model in practice. [IEEE CNSM 2019]
- Integrated SDN and NFV Architecture
 - Led a team of eight researchers to integrate NFV and SDN into a unified architecture for policy-based service deployment and network control. Designed the architecture and its interfaces based on the ETSI NFV framework and prototyped the system using OpenStack, OpenBaton, and OpenDaylight.
- System Backup and Recovery
 - Designed and implemented a Debian-based update and recovery system with automatic failure detection and image rollback using a modified kernel, initrd, and Btrfs file system. On each boot, the system verifies its state and restores a previous working snapshot if a failure is detected.

Teaching and Assistantships

2024	TA, Systems Security, Washington University in St. Louis
2018	Instructor, Computer Networks, Amirkabir University
2017	TA, Switches and Routers Architecture, Amirkabir University
2016-2018	TA, Optimization and Its Applications in Computer Networks, Amirkabir University

Honors and Awards

2022-2027	Washington University Ph.D. scholarship
2016	Winner of National Elite Foundation award for outstanding academic success
2014	Graduated with distinction from M.Sc. program (Ranked 1 st)

Skills

Programming languages: Go, Python, C, Java, BASH, MATLAB, Kotlin, Lua

Technologies (Most recent): Docker, Envoy Proxy, Open Policy Agent, Linux Kernel, Kubernetes, OpenStack, OpenBaton, Ansible, PostgreSQL, CouchDB, Influxdb, gRPC, Prometheus, Grafana

Learning and Optimization: Numerical Optimization, Convex Optimization, Advanced Algorithms, Machine Learning, Deep Learning

References

Professor **Patrick Crowley**, Washington University in St. Louis, pcrowley@wustl.edu

Professor **Raj Jain**, Washington University in St. Louis, jain@wustl.edu